137. $\int_C \mathbf{F} \cdot d\mathbf{r} = -1$

139. 4×10^{29} erg

141. $\int_C \mathbf{F} \cdot d\mathbf{r} \approx 2.9923$

143. circulation $= \pi a^2$ and flux $= 0$

Section 6.4 Exercises

147. $\int_C 2xydx + (x+y)dy = -\frac{8}{3}$

149. $\int_C \sin x \cos ydx + (xy + \cos x \sin y)dy = \frac{1}{12}$

151. $\int_C (-ydx + xdy) = \pi$

153. $\int_C xe^{-2x}dx + \left(x^4 + 2x^2y^2\right)dy = 0$

155. $\int_C y^3dx - x^3ydy = -20\pi$

157. $\int_C -x^2ydx + xy^2dy = 8\pi$

159. $\int_C \left(x^2 + y^2\right)dx + 2xydy = 0$

161. $A = 19\pi$

163. $A = \frac{3\pi}{8}$

165. $\int_{C+} \left(y^2 + x^3\right)dx + x^4dy = 0$

167. $A = \frac{9\pi}{8}$

169. $A = \frac{8\sqrt{3}}{5}$

171. $\int_C \left(x^2y - 2xy + y^2\right)ds = \frac{1}{2}$

173. $\int_C \frac{xdx + ydy}{x^2 + y^2} = 0$

175. $W = \frac{225}{2}$

177. $W = 12\pi$

179. $W = 2\pi$

181. $\int_C y^2dx + x^2dy = \frac{1}{3}$

183. $\int_C \sqrt{1 + x^3}dx + 2xydy = -3$

185. $\int_C \left(3y - e^{\sin x}\right)dx + \left(7x + \sqrt{y^4 + 1}\right)dy = 36\pi$

187. $\int_C \mathbf{F} \cdot d\mathbf{r} = 2$

189. $\int_C (y + x)dx + (x + \sin y)dy = 0$

191. $\int_C xydx + x^3y^3dy = \frac{22}{21}$

193. $\int_C \mathbf{F} \cdot d\mathbf{r} = \frac{15\pi}{4}$

195. $\int_C \sin(x + y)dx + \cos(x + y)dy = 4$

197. $\int_C \mathbf{F} \cdot d\mathbf{r} = \pi$

199. $\int_C \mathbf{F} \cdot \widehat{\mathbf{N}}ds = 4$

201. $\int_C \mathbf{F} \cdot \mathbf{N} ds = 0$

203. $\int_C \left[-y^3 + \sin(xy) + xy\cos(xy)\right] dx + \left[x^3 + x^2\cos(xy)\right] dy = 4.7124$

205. $\int_C \left(y + e^{\sqrt{x}}\right) dx + \left(2x + \cos\left(y^2\right)\right) dy = \frac{1}{3}$

Section 6.5 Exercises

207. False

209. True

211. True

213. $\text{curl}\,\mathbf{F} = \mathbf{i} + x^2\mathbf{j} + y^2\mathbf{k}$

215. $\text{curl}\,\mathbf{F} = \left(xz^2 - xy^2\right)\mathbf{i} + \left(x^2y - yz^2\right)\mathbf{j} + \left(y^2z - x^2z\right)\mathbf{k}$

217. $\text{curl}\,\mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

219. $\text{curl}\,\mathbf{F} = -y\mathbf{i} - z\mathbf{j} - x\mathbf{k}$

221. $\text{curl}\,\mathbf{F} = \mathbf{0}$

223. $\text{div}\,\mathbf{F} = 3yz^2 + 2y\sin z + 2xe^{2z}$

225. $\text{div}\,\mathbf{F} = 2(x + y + z)$

227. $\text{div}\,\mathbf{F} = \frac{1}{\sqrt{x^2+y^2}}$

229. $\text{div}\,\mathbf{F} = a + b$

231. $\text{div}\,\mathbf{F} = x + y + z$

233. Harmonic

235. $\text{div}\,(\mathbf{F} \times \mathbf{G}) = 2z + 3x$

237. $\text{div}\,\mathbf{F} = 2\left(x^2 + y^2 + z^2\right)$

239. $\text{curl}\,\mathbf{r} = \mathbf{0}$

241. $\text{curl}\,\frac{\mathbf{r}}{r^3} = \mathbf{0}$

243. $\text{curl}\,\mathbf{F} = \frac{2x}{x^2+y^2}\mathbf{k}$

245. $\text{div}\,\mathbf{F} = 0$

247. $\text{div}\,\mathbf{F} = 2 - 2e^{-6}$

249. $\text{div}\,\mathbf{F} = 0$

251. $\text{curl}\,\mathbf{F} = \mathbf{j} - 3\mathbf{k}$

253. $\text{curl}\,\mathbf{F} = 2\mathbf{j} - \mathbf{k}$

255. $a = 3$

257. **F** is conservative.

259. $\text{div}\,\mathbf{F} = \cosh x + \sinh y - xy$

261. $(bz - cy)\mathbf{i} + (cx - az)\mathbf{j} + (ay - bx)\mathbf{k}$

263. $\text{curl}\,\mathbf{F} = 2\omega$

265. $\mathbf{F} \times \mathbf{G}$ does not have zero divergence.

267. $\nabla \cdot \mathbf{F} = -200k\left[1 + 2\left(x^2 + y^2 + z^2\right)\right]e^{-x^2+y^2+z^2}$

Section 6.6 Exercises

269. True

271. True

273. $\mathbf{r}(u, v) = \langle u, v, 2 - 3u + 2v\rangle$ for $-\infty \le u < \infty$ and $-\infty \le v < \infty$.

275. $\mathbf{r}(u, v) = \left\langle u, v, \frac{1}{3}(16 - 2u + 4v)\right\rangle$ for $|u| < \infty$ and $|v| < \infty$.

277. $\mathbf{r}(u, v) = \langle 3\cos u, 3\sin u, v\rangle$ for $0 \le u \le \frac{\pi}{2}, 0 \le v \le 3$

279. $A = 28\pi = 87.9646$

281. $\iint_S z dS = 8\pi$

283. $\iint_S \left(x^2 + y^2\right) z dS = 16\pi$

285. $\iint_S \mathbf{F} \cdot \mathbf{N} dS = \frac{4\pi}{3}$

287. $m \approx 13.0639$

289. $m \approx 228.5313$

291. $\iint_S g dS = 3\sqrt{14}$

293. $\iint_S \left(x - y^2 + z\right) dS \approx 0.9617$

295. $\iint_S \left(x^2 + y^2\right) dS = \frac{4\pi}{3}$

297. $\iint_S x^2 z dS = \frac{1023\pi\sqrt{2}}{5}$

299. $\iint_S (z + y)\, dS \approx 10.1$

301. $m = \pi a^3$

303. $\iint_S \mathbf{F} \cdot \mathbf{N} dS = \dfrac{13}{24}$

305. $\iint_S \mathbf{F} \cdot \mathbf{N} dS = \dfrac{3}{4}$

307. $\int_0^8 \int_0^6 \left(4 - 3y + \dfrac{1}{16}y^2 + z\right)\left(\dfrac{1}{4}\sqrt{17}\right) dzdy$

309. $\int_0^2 \int_0^6 \left[x^2 - 2(8 - 4x) + z\right] \sqrt{17} dzdx$

311. $\iint_S \left(x^2 z + y^2 z\right) dS = \frac{\pi a^5}{2}$

313. $\iint_S x^2 yz dS = 171\sqrt{14}$

315. $\iint_S yz dS = \frac{\sqrt{2}\pi}{4}$

317. $\iint_S (x\mathbf{i} + y\mathbf{j}) \cdot dS = 16\pi$

319. $m = \frac{\pi a^7}{192}$

321. $F \approx 4.57$ lb.

323. $8\pi a$

325. The net flux is zero.

Section 6.7 Exercises

327. $\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{N}) dS = \pi a^2$

329. $\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{N}) dS = 18\pi$

331. $\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{N}) dS = -8\pi$

333. $\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{N}) dS = 0$

335. $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$

337. $\int_s \mathbf{F} \cdot d\mathbf{r} = -3\pi \approx -9.4248$

339. $\iint_S \text{curl}\, \mathbf{F} \cdot d\mathbf{S} = 0$

341. $\iint_S \text{curl}\, \mathbf{F} \cdot d\mathbf{S} = 2.6667$

343. $\iint_S (\text{curl}\, \mathbf{F} \cdot \mathbf{N}) dS = -\frac{1}{6}$

345. $\int\limits_C \left(\frac{1}{2}y^2\,dx + z\,dy + x\,dz\right) = -\frac{\pi}{4}$

347. $\iint_S (\text{curl}\,\mathbf{F}\cdot\mathbf{N})dS = 3\pi$

349. $\int_C (c\mathbf{k}\times\mathbf{R})\cdot d\mathrm{r} = 2\pi c$

351. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 0$

353. $\int_C \mathbf{F}\cdot d\mathbf{r} = -4$

355. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 0$

357. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = -36\pi$

359. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 0$

361. $\int_C \mathbf{F}\cdot d\mathbf{r} = 0$

363. $\iint_S \text{curl}(\mathbf{F})\cdot d\mathbf{S} = 84.8230$

365. $A = \iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 0$

367. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 2\pi$

369. $C = \pi(\cos\varphi - \sin\varphi)$

371. $\int_C \mathbf{F}\cdot d\mathbf{r} = 48\pi$

373. $\iint_S \text{curl}\,\mathbf{F}\cdot d\mathbf{S} = 0$

375. 0

Section 6.8 Exercises

377. $\int_S \mathbf{F}\cdot\mathbf{N}ds = 24\pi \approx 75.3982$

379. $\int_S \mathbf{F}\cdot\mathbf{N}ds = \frac{243\pi}{2} \approx 381.704$

381. $\int_S \mathbf{F}\cdot\mathbf{N}ds = 12\pi \approx 37.6991$

383. $\int_S \mathbf{F}\cdot\mathbf{N}ds = \frac{9\pi a^4}{2}$

385. $\iint_S \mathbf{F}\cdot d\mathbf{S} = \frac{4\pi}{3}$

387. $\iint_S \mathbf{F}\cdot d\mathbf{S} = 0$

389. $\iint_S \mathbf{F}\cdot d\mathbf{S} = \frac{384\pi}{5} \approx 241.2743$

391. $\iint_D \mathbf{F}\cdot d\mathbf{S} =$ Net flux $= 0$; flux through the paraboloid $= -\pi$

393. $\iint_S \mathbf{F}\cdot d\mathbf{S} = \frac{2\pi}{3}$

395. $16\sqrt{6}\pi$

397. $-\frac{128\pi}{3}$

399. $-224\pi \approx -703.7168$

401. 20

403. $\iint_S \mathbf{F}\cdot d\mathbf{S} = 8$

405. $\iint_S \mathbf{F}\cdot\mathbf{N}dS = \frac{1}{8}$

407. $\iint_S \|\mathbf{R}\| \, \mathbf{R} \cdot \mathbf{n} dS = 4\pi a^4$

409. $\iiint_R z^2 dV = \frac{4\pi}{15}$

411. $\iint_S \mathbf{F} \cdot d\mathbf{S} = 3e - 2 + \frac{1}{2}\sin 1 \approx 6.5759$

413. $\iint_S \mathbf{F} \cdot d\mathbf{S} = 21$

415. $\iint_S \mathbf{F} \cdot d\mathbf{S} = 72$

417. $\iint_S \mathbf{F} \cdot d\mathbf{S} = -\frac{32\pi}{3} \approx -33.5103$

419. $\iint_S \mathbf{F} \cdot d\mathbf{S} = \pi a^4 b^2$

421. $\iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{5\pi}{2}$

423. $\iint_S \mathbf{F} \cdot d\mathbf{S} = \frac{21\pi}{2}$

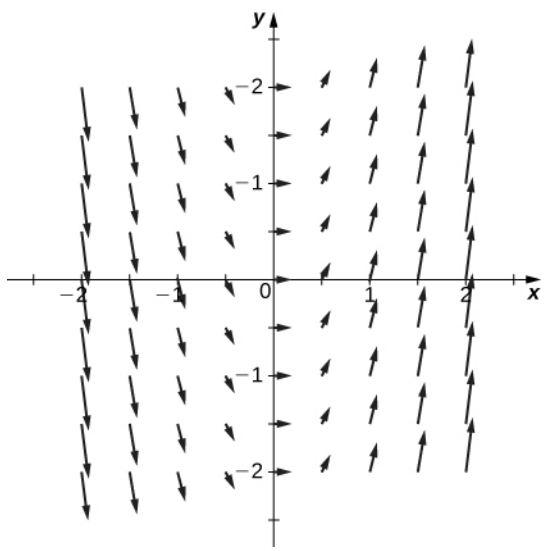
425. $e^{-1} - 1$

Review Exercises

427. False

429. False

431.



433. Conservative, $f(x, y) = xy - 2e^y$

435. Conservative, $f(x, y, z) = x^2 y + y^2 z + z^2 x$

437. $-\frac{16}{3}$

439. $\frac{32\sqrt{2}}{9}\left(3\sqrt{3} - 1\right)$

441. Divergence: $e^x + xe^{xy} + xye^{xyz}$, curl: $xze^{xyz}\mathbf{i} - yze^{xyz}\mathbf{j} + ye^{xy}\mathbf{k}$

443. -18π

445. $-\pi$

447. 24π

449. $\sqrt{2}\left(2\sqrt{2} + \pi\right) = 4 + \pi\sqrt{2}$

451. $8\pi/3$

Chapter 7

Checkpoint

7.1 a. Nonlinear
b. Linear, nonhomogeneous

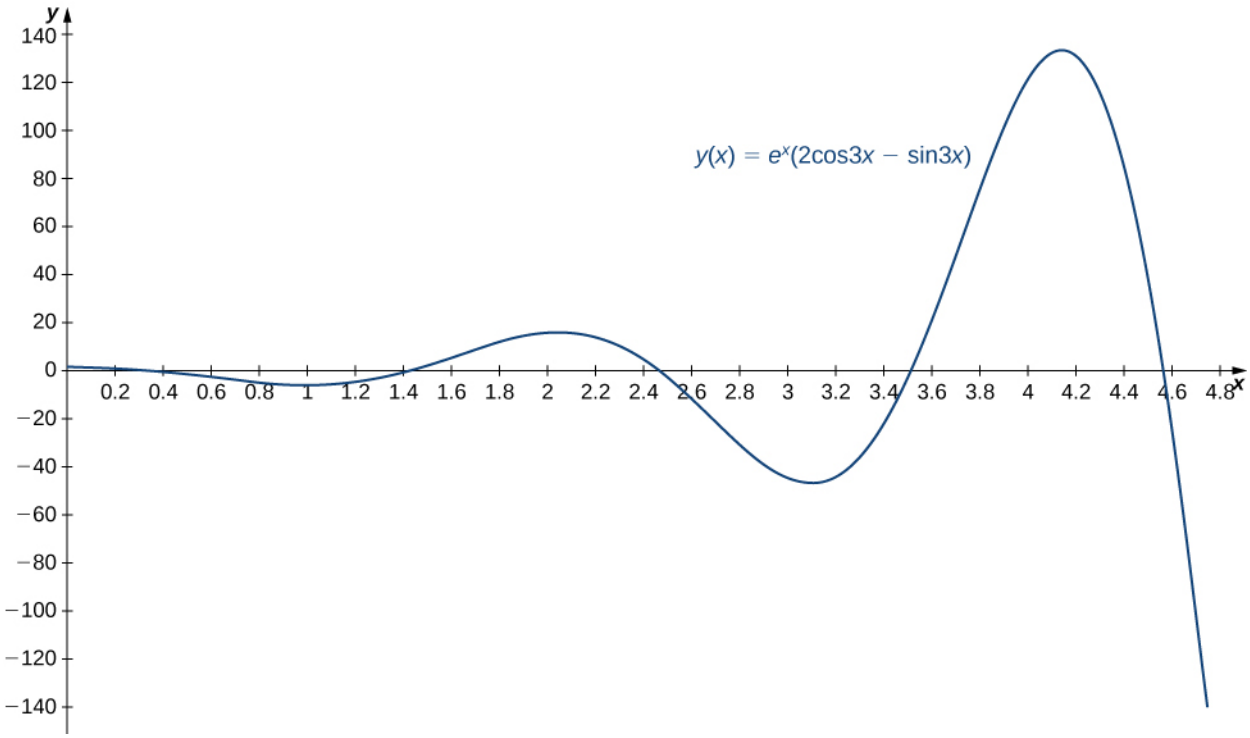
7.4 Linearly independent

7.5 $y(x) = c_1 e^{3x} + c_2 x e^{3x}$

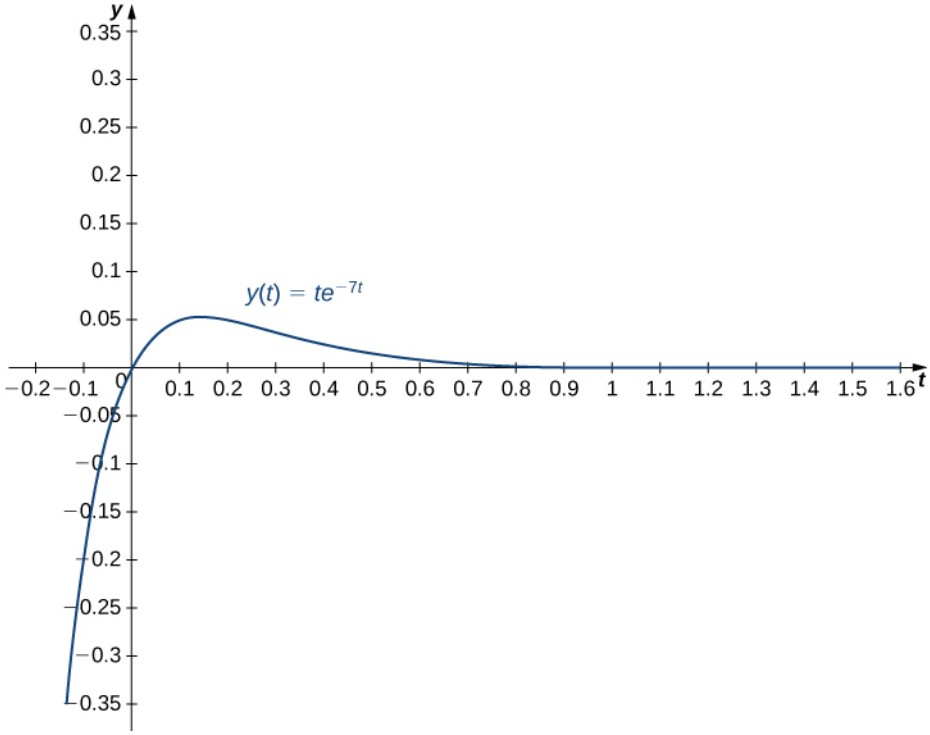
7.6 a. $y(x) = e^x (c_1 \cos 3x + c_2 \sin 3x)$
b. $y(x) = c_1 e^{-7x} + c_2 x e^{-7x}$

7.7 $y(x) = -e^{-2x} + e^{5x}$

7.8 $y(x) = e^x (2\cos 3x - \sin 3x)$



7.9 $y(t) = te^{-7t}$



At time $t = 0.3$, $y(0.3) = 0.3e^{(-7*0.3)} = 0.3e^{-2.1} \approx 0.0367$. The mass is 0.0367 ft below equilibrium. At time $t = 0.1$, $y'(0.1) = 0.3e^{-0.7} \approx 0.1490$. The mass is moving downward at a speed of 0.1490 ft/sec.

7.10 $y(x) = c_1 e^{-x} + c_2 e^{4x} - 2$

7.11 $y(t) = c_1 e^{2t} + c_2 te^{2t} + \sin t + \cos t$

7.12 a. $y(x) = c_1 e^{4x} + c_2 e^{x} - xe^{x}$
b. $y(t) = c_1 e^{-3t} + c_2 e^{2t} - 5\cos 2t + \sin 2t$

7.13 $z_1 = \frac{3x+3}{11x^2}$, $z_2 = \frac{2x+2}{11x}$

7.14 a. $y(x) = c_1 \cos x + c_2 \sin x + \cos x \ln|\cos x| + x \sin x$
b. $x(t) = c_1 e^{t} + c_2 te^{t} + te^{t} \ln|t|$

7.15 $x(t) = 0.1\cos(14t)$ (in meters); frequency is $\frac{14}{2\pi}$ Hz.

7.16 $x(t) = \sqrt{17}\sin(4t + 0.245)$, frequency $= \frac{4}{2\pi} \approx 0.637$, $A = \sqrt{17}$

7.17 $x(t) = 0.6e^{-2t} - 0.2e^{-6t}$

7.18 $x(t) = \frac{1}{2}e^{-8t} + 4te^{-8t}$

7.19 $x(t) = -0.24e^{-2t}\cos(4t) - 0.12e^{-2t}\sin(4t)$

7.20 $x(t) = -\frac{1}{2}\cos(4t) + \frac{9}{4}\sin(4t) + \frac{1}{2}e^{-2t}\cos(4t) - 2e^{-2t}\sin(4t)$
Transient solution: $\frac{1}{2}e^{-2t}\cos(4t) - 2e^{-2t}\sin(4t)$
Steady-state solution: $-\frac{1}{2}\cos(4t) + \frac{9}{4}\sin(4t)$

7.21 $q(t) = -25e^{-t}\cos(3t) - 7e^{-t}\sin(3t) + 25$

7.22 a. $y(x) = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n} = a_0 e^{-x^2}$
b. $y(x) = a_0 (x+1)^3$

Section 7.1 Exercises

1. linear, homogenous

3. nonlinear

5. linear, homogeneous

11. $y = c_1 e^{5x} + c_2 e^{-2x}$

13. $y = c_1 e^{-2x} + c_2 xe^{-2x}$

15. $y = c_1 e^{5x/2} + c_2 e^{-x}$

17. $y = e^{-x/2}\left(c_1 \cos \frac{\sqrt{3}x}{2} + c_2 \sin \frac{\sqrt{3}x}{2}\right)$

19. $y = c_1 e^{-11x} + c_2 e^{11x}$

21. $y = c_1 \cos 9x + c_2 \sin 9x$

23. $y = c_1 + c_2 x$

25. $y = c_1 e^{\left(\left(1+\sqrt{22}\right)/3\right)x} + c_2 e^{\left(\left(1-\sqrt{22}\right)/3\right)x}$

27. $y = c_1 e^{-x/6} + c_2 x e^{-x/6}$

29. $y = c_1 + c_2 e^{9x}$

31. $y = -2e^{-2x} + 2e^{-3x}$

33. $y = 3\cos(2x) + 5\sin(2x)$

35. $y = -e^{6x} + 2e^{-5x}$

37. $y = 2e^{-x/5} + \frac{7}{5} x e^{-x/5}$

39. $y = \left(\frac{2}{e^6 - e^{-7}}\right) e^{6x} - \left(\frac{2}{e^6 - e^{-7}}\right) e^{-7x}$

41. No solutions exist.

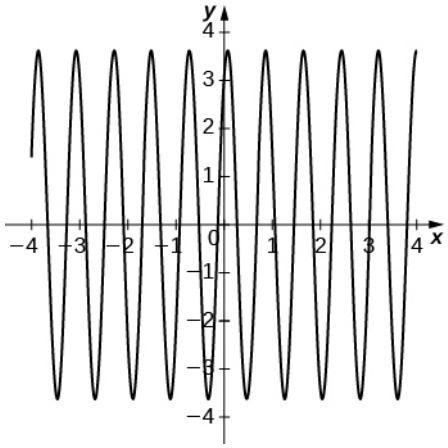
43. $y = 2e^{2x} - \frac{2e^2+1}{e^2} x e^{2x}$

45. $y = 4\cos 3x + c_2 \sin 3x$, infinitely many solutions

47. $5y'' + 19y' - 4y = 0$

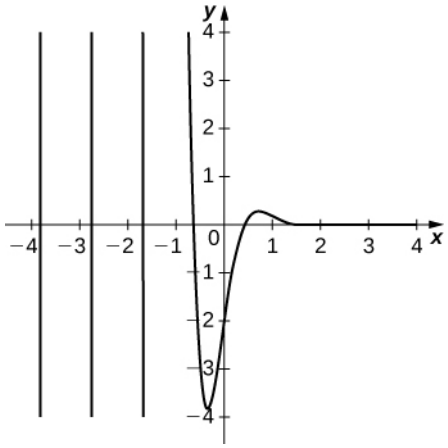
49. a. $y = 3\cos(8x)$

b.



51. a. $y = e^{(-5/2)x}\left[-2\cos\left(\frac{\sqrt{35}}{2}x\right) + \frac{4\sqrt{35}}{35}\sin\left(\frac{\sqrt{35}}{2}x\right)\right]$

b.



Section 7.2 Exercises

55. $y = c_1 e^{-4x/3} + c_2 e^x - 2$

57. $y = c_1 \cos 4x + c_2 \sin 4x + \frac{1}{20} e^{-2x}$

59. $y = c_1 e^{2x} + c_2 x e^{2x} + 2x^2 + 5x + 4$

61. $y = c_1 e^{-x} + c_2 x e^{-x} + \frac{1}{2} \sin x - \frac{1}{2} \cos x$

63. $y = c_1 \cos x + c_2 \sin x - \frac{1}{3} x \cos 2x - \frac{5}{9} \sin 2x$

65. $y = c_1 e^{-5x} + c_2 x e^{-5x} + \frac{1}{6} x^3 e^{-5x} + \frac{4}{25}$

67. a. $y_p(x) = Ax^2 + Bx + C$

b. $y_p(x) = -\frac{1}{3} x^2 + \frac{4}{3} x - \frac{35}{9}$

69. a. $y_p(x) = \left(Ax^2 + Bx + C\right) e^{-x}$

b. $y_p(x) = \left(\frac{1}{4} x^2 - \frac{5}{8} x - \frac{33}{32}\right) e^{-x}$

71. a. $y_p(x) = \left(Ax^2 + Bx + C\right) e^x \cos x + \left(Dx^2 + Ex + F\right) e^x \sin x$

b. $y_p(x) = \left(-\frac{1}{10} x^2 - \frac{11}{25} x - \frac{27}{250}\right) e^x \cos x + \left(-\frac{3}{10} x^2 + \frac{2}{25} x + \frac{39}{250}\right) e^x \sin x$

73. $y = c_1 + c_2 e^{-2x} + \frac{1}{15} e^{3x}$

75. $y = c_1 e^{2x} + c_2 e^{-4x} + x e^{2x}$

77. $y = c_1 e^{3x} + c_2 e^{-3x} - \frac{8x}{9}$

79. $y = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{2} x \cos 2x + \frac{3}{4} \sin 2x \ln(\sin 2x)$

81. $y = -\frac{347}{343} + \frac{4}{343} e^{7x} + \frac{2}{7} x^2 e^{7x} - \frac{4}{49} x e^{7x}$

83. $y = -\frac{57}{25} + \frac{3}{25} e^{5x} + \frac{1}{5} x e^{5x} + \frac{4}{25} e^{-5x}$

85. $y_p = \frac{1}{2} + \frac{10}{3} x^2 \ln x$

Section 7.3 Exercises

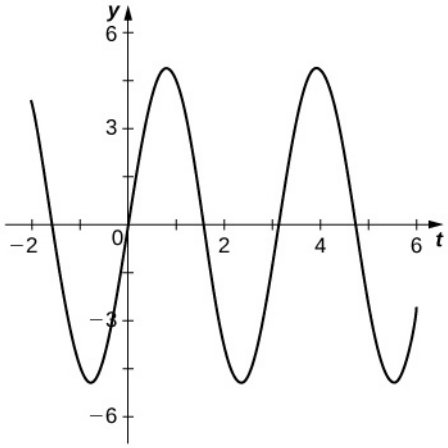
87. $x'' + 16x = 0$, $x(t) = \frac{1}{6} \cos(4t) - 2 \sin(4t)$, period $= \frac{\pi}{2}$ sec, frequency $= \frac{2}{\pi}$ Hz

89. $x'' + 196x = 0$, $x(t) = 0.15 \cos(14t)$, period $= \frac{\pi}{7}$ sec, frequency $= \frac{7}{\pi}$ Hz

91. a. $x(t) = 5 \sin(2t)$

b. period $= \pi$ sec, frequency $= \frac{1}{\pi}$ Hz

c.



d. $t = \frac{\pi}{2}$ sec

93. a. $x(t) = e^{-t/5} (20 \cos(3t) + 15 \sin(3t))$

b. underdamped

95. a. $x(t) = e^{-4t} \left(-5 \cos 11.866t - 0.834 \sin 11.866t\right)$

b. underdamped

97. $x(\pi) = \frac{7e^{-\pi/4}}{6}$ ft below

99. $x\left(t\right) = \frac{1}{2} \cos\left(8\sqrt{2}t\right) - \frac{4\sqrt{2}}{21} \sin\left(8\sqrt{2}\mathrm{t}\right) + \frac{8}{21} \sin\left(8\sqrt{2}\mathrm{t}\right)$

101. $q(t) = e^{-6t}\left(0.051\cos(8t) + 0.03825\sin(8t)\right) - \frac{1}{20}\cos(10t)$

103. $q(t) = e^{-10t}(-32t - 5) + 5, \quad I(t) = 2e^{-10t}(160t + 9)$

Section 7.4 Exercises

105. $y = a_0 + 5a_1 \sum_{n=1}^{\infty} \frac{(-x/5)^n}{n!} = c_0 + 5c_1 e^{-x/5}$

107. $y = a_0 \sum_{n=0}^{\infty} \frac{(x)^{2n}}{(2n)!} + a_1 \sum_{n=0}^{\infty} \frac{(x)^{2n+1}}{(2n+1)!}$

109. $y = a_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{n!} = c_0 e^{x^2}$

111. $y = a_0 \sum_{n=0}^{\infty} \frac{x^{2n}}{2^n n!} + a_1 \sum_{n=0}^{\infty} \frac{x^{2n+1}}{1 \cdot 3 \cdot 5 \cdot 7 \cdots (2n+1)}$

113. $y = c_0 x^3 + \frac{c_1}{x}$

115. $y = 1 - 3x + \frac{2x^3}{3!} - \frac{12x^4}{4!} + \frac{16x^6}{6!} - \frac{120x^7}{7!} + \cdots$

Review Exercises

117. True

119. False

121. second order, linear, homogeneous, $\lambda^2 - 2 = 0$

123. first order, nonlinear, nonhomogeneous

125. $y = c_1\cos(3x) + c_2\sin(3x)$

127. $y = c_1 e^x\cos(3x) + c_2 e^x\sin(3x) + \frac{2}{5}x + \frac{2}{25}$

129. $y = c_1 e^{-x} + c_2 e^{-4x} + \frac{x}{4} + \frac{e^{2x}}{18} - \frac{5}{16}$

131. $y = c_1 e^{(-3/2)x} + c_2 x e^{(-3/2)x} + \frac{4}{9}x^2 + \frac{4}{27}x - \frac{16}{27}$

133. $y = e^{-2x}\sin\left(\sqrt{2}x\right)$

135. $y = \frac{e^{1-x}}{e^4-1}\left(e^{4x} - 1\right)$

137. $\theta(t) = \theta_0\cos\left(\sqrt{\frac{g}{l}}L\right)$

141. $b = \sqrt{a}$

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